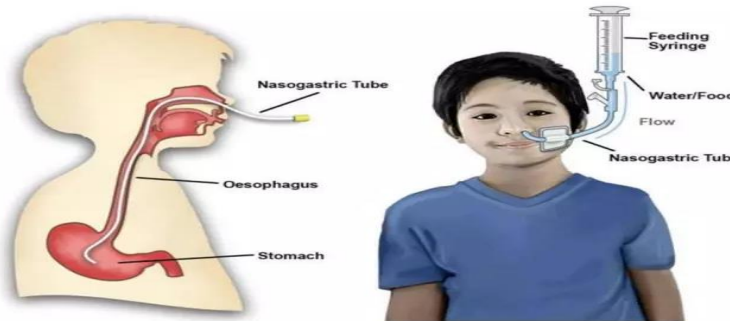


Nutrition Support to the Critically ill Patients



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Outline

- ▶ **Introduction**
- ▶ **Definition of Malnutrition in critically-ill patients**
- ▶ **General goals of nutritional support**
- ▶ **Methods Of Nutritional Support**
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- ▶ **Nursing Role In Nutritional Care of Critically ill Patients**



Introduction

Nutrition is recognized as a vital component of health so that an optimum nutritional status helps maintain health, promotes normal growth and development, regulates body process and protects the body against diseases.

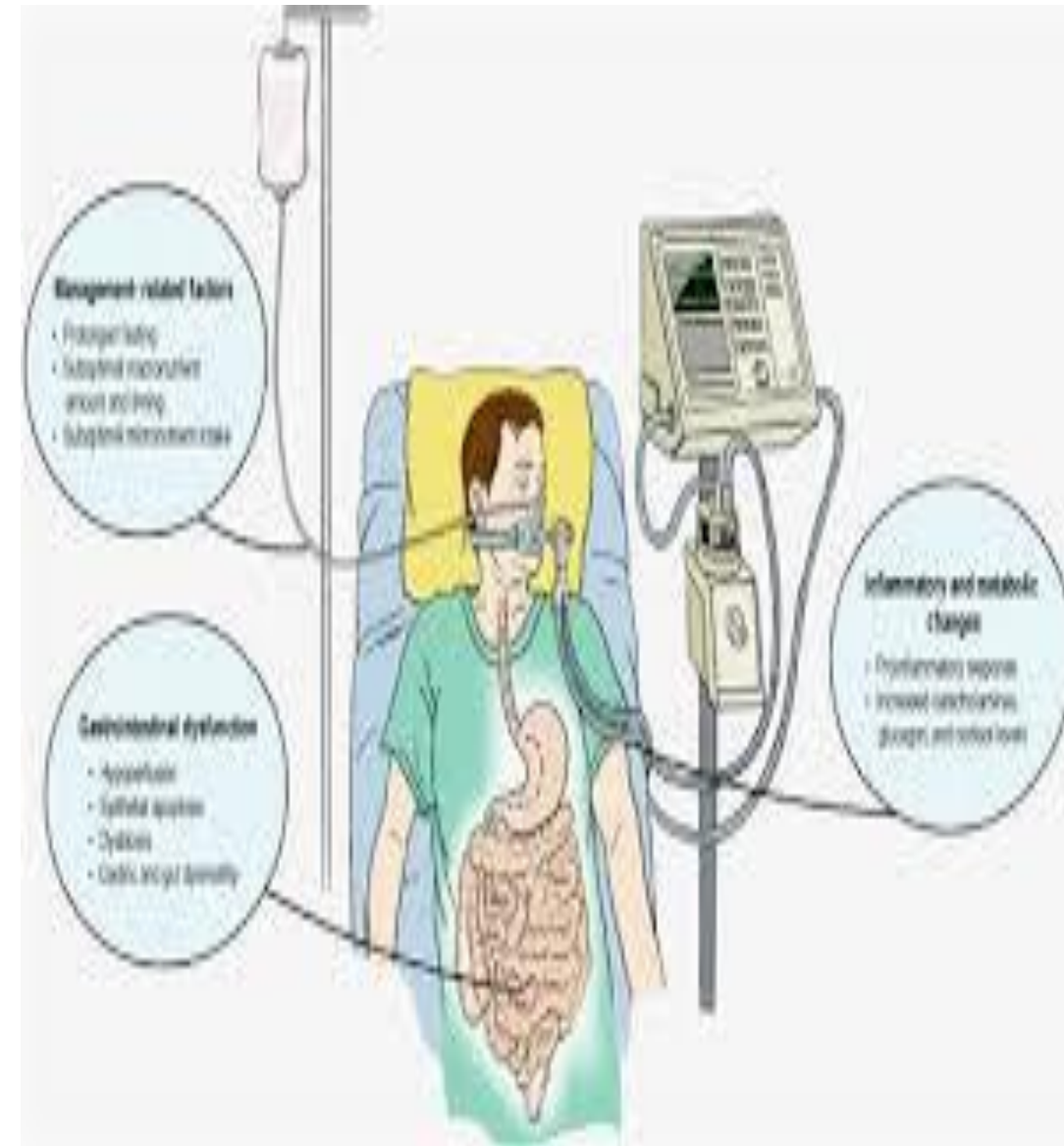


Critically ill patients are **hypermetabolic** and **have increased nutrient requirements**. So that Nutritional support has become a **routine part** of the care of critically ill patients



Definition malnutrition in critically-ill patients:

Refers to a generalized disordering of body composition such that either macro- or micro- nutrient intake is sub-optimal. Severe deficits can result in **organ dysfunction**, **abnormalities in blood biochemistry** and **reduced body mass**.



Causes of malnutrition in A critically ill patient

❑ **Exposed to stress** are lead to more damaging because protein becomes more significant energy substrate.
stress in form as

- Sepsis
- Multiple organ system failure
- Respiratory compromise.
- Major burn,
- Traumatic injury, surgery

❑ **Other cause**

- Limited nutrient intake
- Increased caloric losses,
- Increased catabolic state are often associated with acute severe illness and frequent presence of prior chronic wasting condition.
- limited absorption,
- Increased nutrient requirement

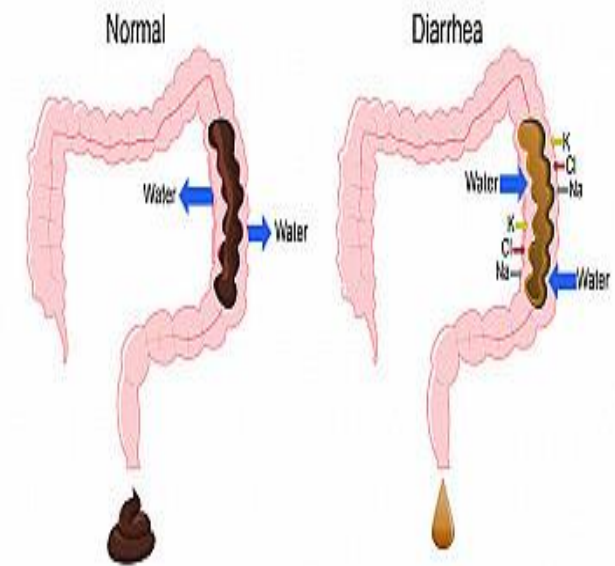
Causes of malnutrition in A critically ill patient

- ❖ **Contributing factors affecting absorption in the critically ill that include:** intestinal villous atrophy which results in a reduction in the surface area available for absorption; digestive enzyme deficiency related to malnutrition.
- ❖ **Increased losses** are another factor which may contribute to malnutrition in the critically ill patient. **Diarrhea** is considered the most important cause for increasing loss in ICU patient.

The Causes of diarrhea are:

- **Prolonged bowel rest** and its subsequent gut atrophy,
- **Lack of fiber** in many feeding product can contribute to diarrhea as fiber normalizes stool pattern and provides a source of short chain fatty acids that aiding in the prevention of diarrhea, Also Deficiencies in vitamin A can contribute to the development of diarrhea.
- **Also hypertonic tube feeding** is often implicated as a cause of diarrhea and both the osmolality of the enteral product.

Diarrhea



The Causes of diarrhea are:

- **Hypoalbuminemia** can contribute to the development of diarrhea in critically ill. Colloid osmotic pressure decreases with Hypoalbuminemia and results in mucosal edema and impaired absorption. Albumin level of less than 2.5 g/dl has been reported to increase the likelihood of diarrhea.
- ❖ **Stress ulcer** related GI bleeding is another serious complication which increases morbidity and mortality in ICU. H2 (histamine) antagonists and antacids are still prescribed to reduce stress ulceration and bleeding. The buffering of hydrochloric acid reduces the bacteriostatic capabilities of the stomach.

General goals of nutritional support:

- To provide patients with their general nutrient requirements

❖ Nutrient requirements depend on

- ✓ Individual patient needs
- ✓ Specific disease processes

The Harris-Benedict equation can be used or even more simply, requirements can be guessed **25 total kilocalories /kg/day** is the estimate most often used.

Calories can be given in three forms:

- ▶ **Carbohydrate:** 30 to 70% of the total calories can be given as carbohydrate.
- ▶ **Fat:** 20 to 50% of the total calories can be given as fat.
- ▶ **Protein:** 15 to 20% of the total calories per day can be administered as protein or amino acids depending on route of administration.
- **Micronutrient** Approximately 1mmol/kg of both **sodium and potassium**
- **Other electrolytes**, e.g. magnesium, iron, copper, zinc and selenium are also necessary, but in much smaller amounts.
- **Other micronutrients include**
 - ✓ fat soluble vitamins (vitamins A, D, E , K and carotene)
 - ✓ water-soluble vitamins (B, C,).



Fluid requirements

- **Factors are affected Fluid needs by**

- Functional cardiac
- Pulmonary
- Hepatic
- Renal status

Fluid requirements increase with

- fever
- Diarrhea
- Hemorrhage
- Surgical drains
- Loss of skin integrity (ie, burns, open wounds).

- **Whereas patients with cardiac, pulmonary or renal disease may require less fluid intake.**
- The average adult requires approximately **35-45 ml/kg of water per day**,



Fluid needs are determined by careful monitoring, particularly:



Patient's physical appearance

Dry mucous membranes with **volume depletion** - Edema with **volume excess**

- **Serum sodium concentrations**

- Hyponatremia may occur in the presence of excess, normal or depleted fluid volume.
- Hypernatremia may result from pure water loss, pure salt gain or hypotonic fluid loss.

- **BUN : Creatinine ratio**

- Normal BUN : SrCr ratio is 10:1 to 15:1. - Renal failure or abnormalities in fluid volume may alter this ratio.

- **Intake and output**

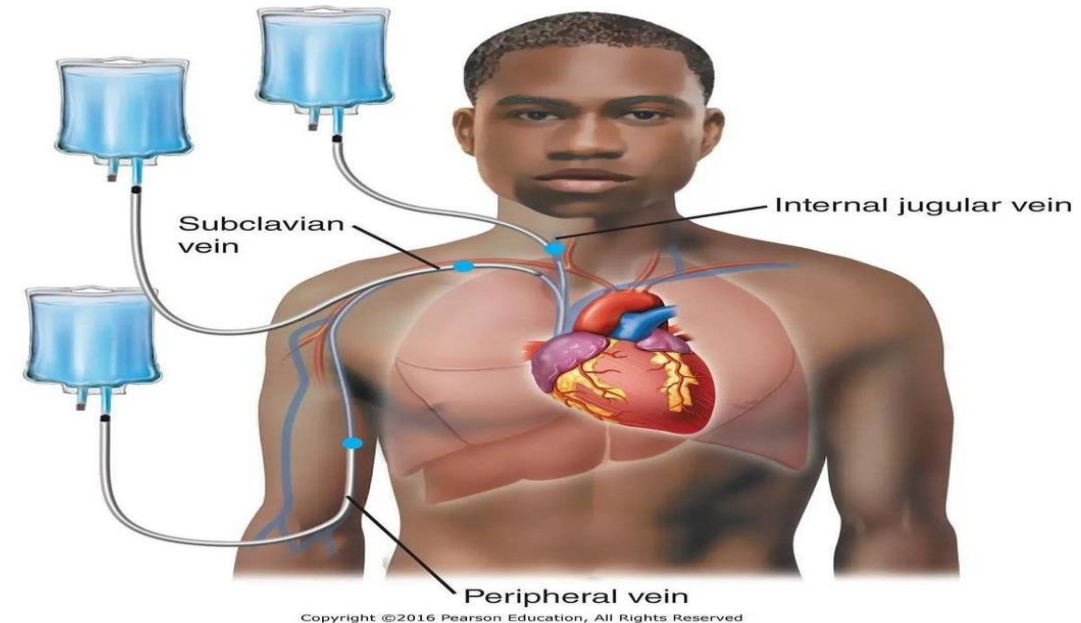
- Urine output - Other fluid losses (vomiting, diarrhea, skin losses and fistula drainages)

Methods Of Nutritional Support:

□ Enteral Nutrition

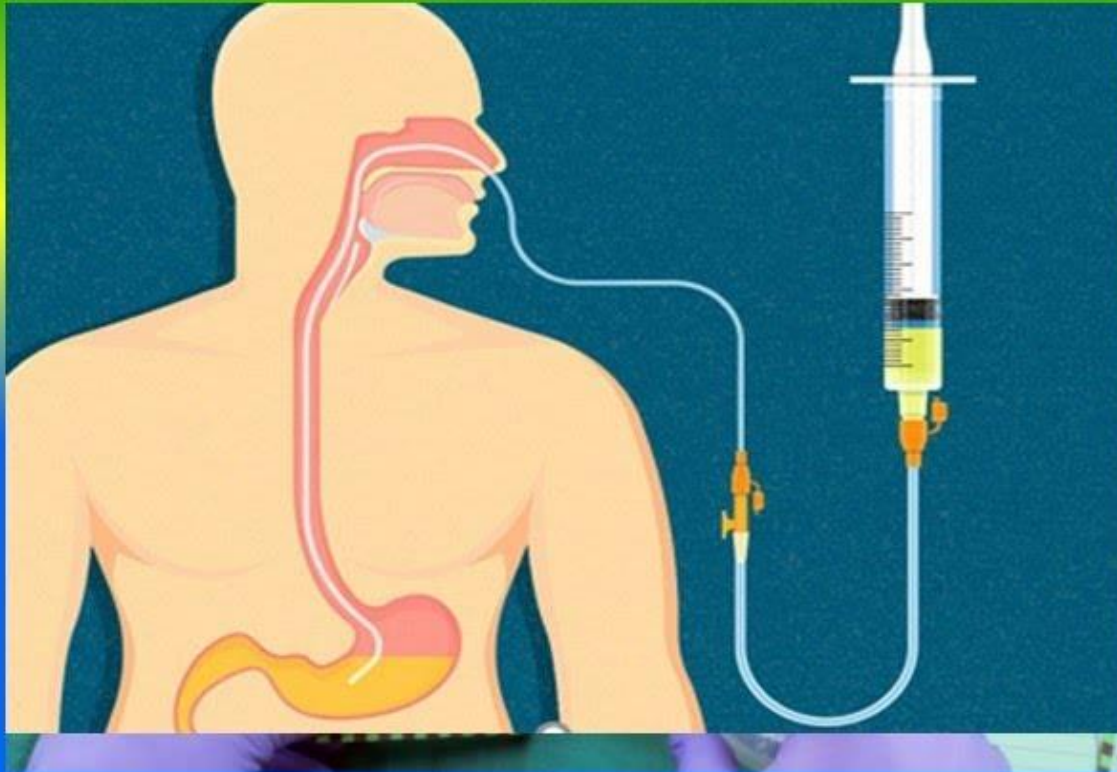


□ Parenteral Nutrition

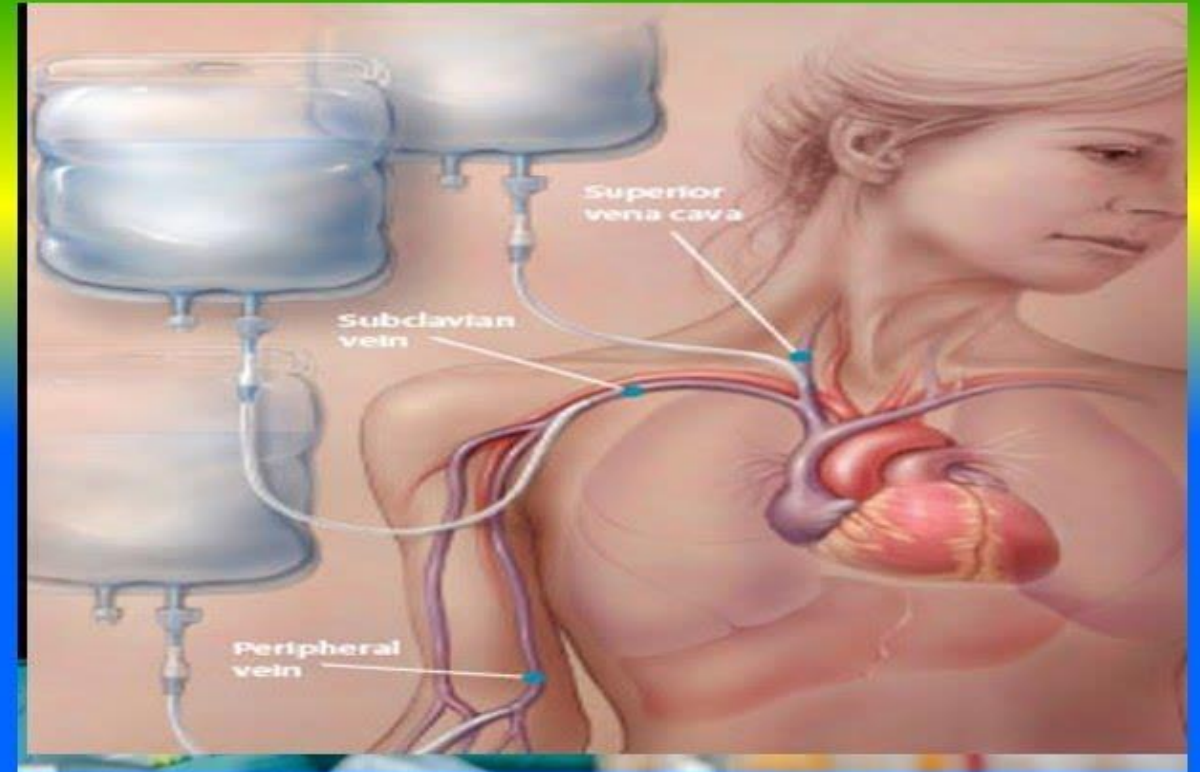


Difference Between

Enteral Nutrition



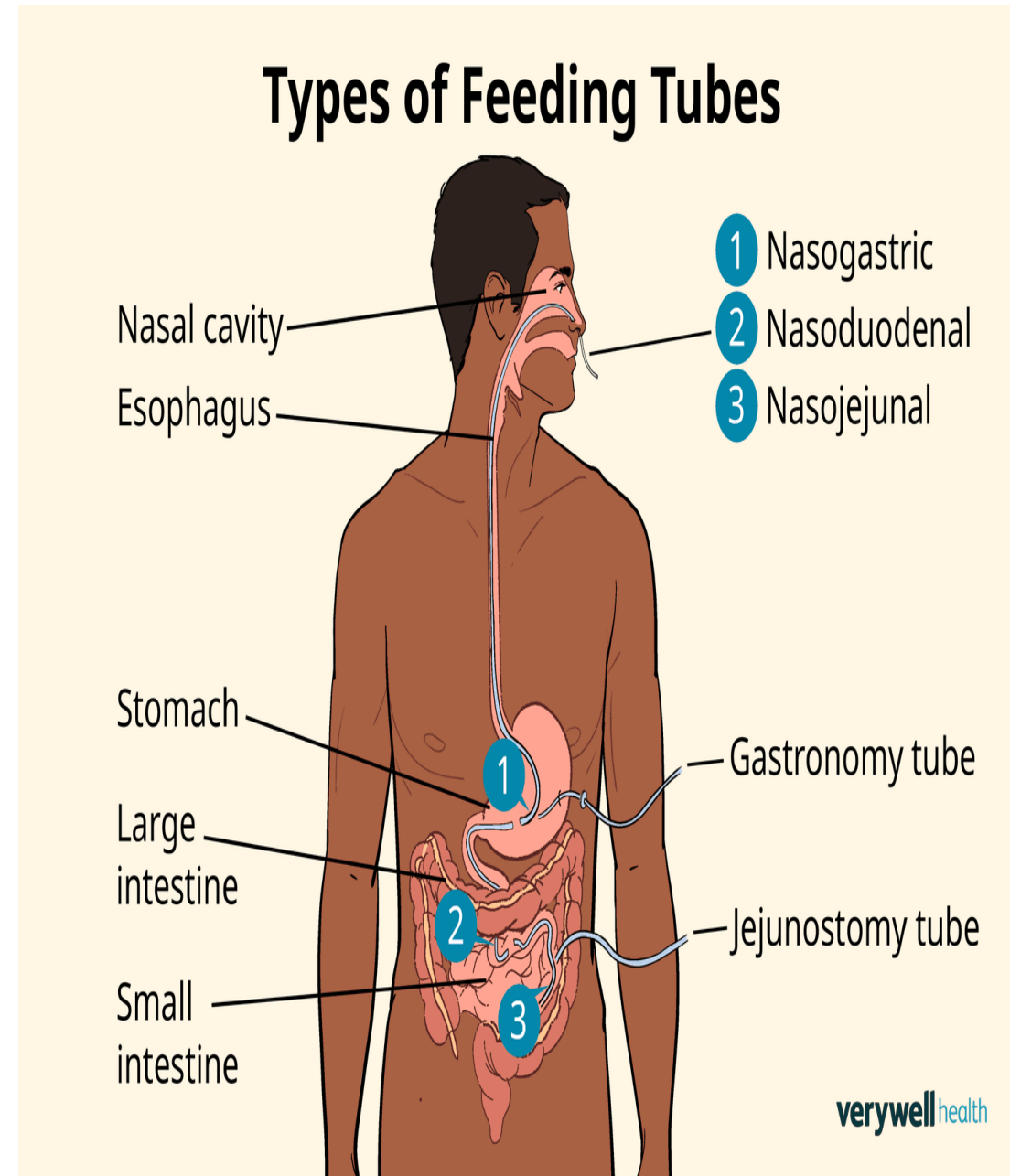
Parenteral Nutrition



Enteral Nutrition

defined, as nutrition via the **gastro-intestinal (GI) tract** but generally is interpreted as nutrition via **tube feeding** into GI tract through

- Orogastric
- Esophagostomy
- Gastrostomy
- Nasogastric
- Pharyngotomy
- Jejunostomy tubes.



The rational for prescribing enteral nutrition rather than parenteral nutrition is based on the following:

(1) Maintenance of gut structure and function

(2) Enhanced use of nutrients

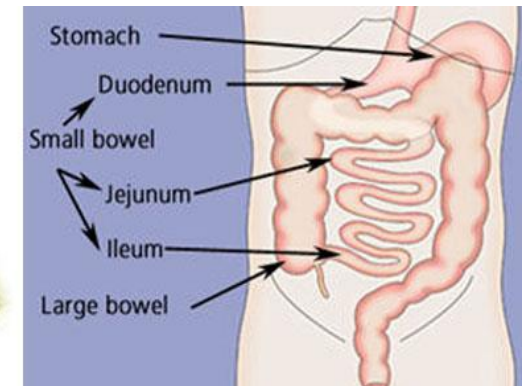
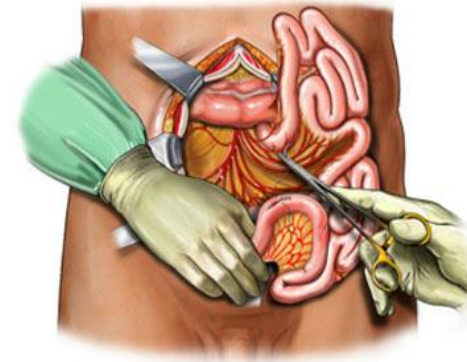
(3) Safety of administration

(4) Reduced cost.

(5) Enteral nutrition is considered safer than parenteral nutrition when it is administered carefully

Indication enteral nutrition

- **Functional GI tracts** who are unable to consume adequate amounts of energy (Kcals) and or nutrients by mouth.
- **Protein-calorie malnutrition** with inadequate oral intake for the previous 5 days, normal nutritional status with less than 50% of required oral intake for the previous 7-10 days,
- **Severe dysphagia**
- **Major full-thickness burn,**
- **Massive small-bowel resection** in combination with administration of total parental nutrition.
- Major trauma,
- Radiation therapy, mild chemotherapy
- Liver or kidney failure.



Enteral formulas can be categorized as

- ❖ Polymeric formula

- ❖ Monomeric formulas

- ❖ Disease specific formula

- ❖ **The diseases-specific formulas** have been formulated to meet the needs of patients with special disease-related needs. These include renal, hepatic, pulmonary, cardiac, trauma and stress and clear liquid formulas

Nutrients in polymeric formulas

- Polymeric formulas range from those made from blenderized foods to those compounded from protein isolates (e.g., caseinate & soy), vegetable oils and carbohydrates. Lactose has been eliminated as a carbohydrate source from most compounded formulas because patients with primary lactase deficiency do not tolerate it.
- Since secondary lactose deficiency may result from critical illness, the critically ill patient is also unlikely to tolerate lactose.



polymeric formulas

- The caloric density of most of these formulas is 1 Kcal per milliliter, although some provide 1.5 or 2 Kcal per milliliter.
- The higher-calorie formulas are useful in meeting elevated energy requirements of the critically ill patient.
- Nurses should observe the patient for tolerance of formula as evidenced by absence of abdominal distension, nausea, vomiting or diarrhea.



Monomeric formulas

- Provide nutrients in simple or "**predigested**" form and very little digestion is required before nutrients can be absorbed.
- These formulas are compounded from nutrient sources ranging from simple **crystalline amino acids** and **glucose molecules**, to slightly more complex sources such as di- and tripeptides and modified corn starch or glucose oligosaccharides.
- Most monomeric formulas are **powder** and **require reconstitution**.
- **Caloric density** depends upon final dilution, but all full strength they provide 1 Kcal per milliliter .



Complications enteral nutrition

1- Gastrointestinal complications include : nausea, vomiting , diarrhea incidence rate ranges from 2.3% to 63% and constipation.

Nausea, vomiting, or gastroparesis can be caused by

- Medications Critical illness
- Rapid or bolus infusion
- Diabetes Neurologic dysfunction
- Improper tube placement. GI infection
- GI dysfunction (hypermotility or malabsorption), and hyperosmolar or low residue formula

Constipation caused by lack of adequate hydration, long-term fiber-free feedings, bed rest, impaction, obstruction, and narcotics .

2- Metabolic complications

- **Dehydration** due to inadequate fluid intake or excessive fluid losses through diarrhea, diuresis, wounds, or fever.
- **Overhydration** during enteral feeding is possible when metabolic demands are high and organ function (Cardiac, renal, or hepatic) is impaired.
- **Hyperglycemia** can occur with enteral nutrition. It is most common in patients with Diabetes, Metabolic stress , Sepsis or with the use of steroids or excessive glucose administration
- **Electrolyte and mineral abnormalities**

3- Pulmonary complications

Pulmonary aspiration are potentially the most dangerous complications associated with enteral feedings and may even result in **death** .

Pulmonary aspiration manifested with

- Fever
- Tachypnea
- Tachycardia
- Hypoxaemia, hypercapnia, and/or new pulmonary infiltrate become evident on X-ray without any other obvious cause.

4-Tube occlusion

Causes of tube occlusion

- ☐ Inadequate tube flushing,
- ☐ Administration of medication through the tube.

The researches added pill fragments formula residue adhering to the tube lumen.

5- Mechanical complication

Can be divided into two types of complications,

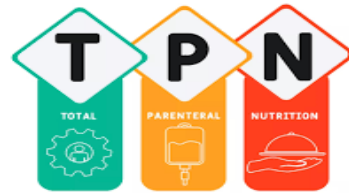
- ❖ **First type** occurs during tube insertion
- ❖ **Second type** occurs during blind insertion of small bore feeding tubes with rigid stylist is relatively common problem. These tubes must be inserted with caution and without undue force.

Prevent enteral nutrition complication

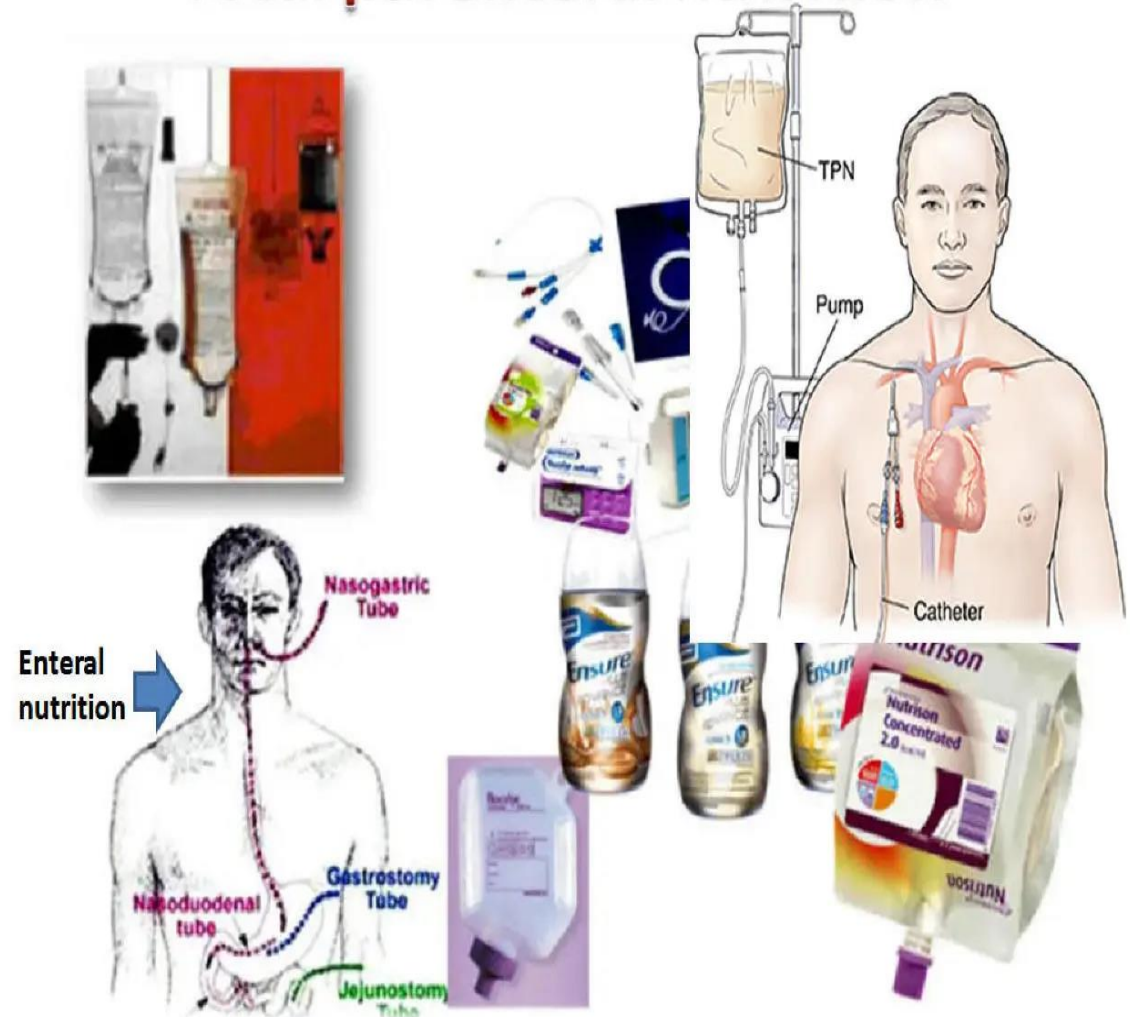
- Close monitoring , timely and accurate assessment of a patient's tolerance to feeding.
- The clinical nurse plays a vital role in providing these observations and can assist easily in complication prevention.
- Clinical pathways are useful tools that allow the clinicians to track patient outcomes related to enteral therapy and to document variances as they occur.

Parenteral Nutrition

Total parenteral nutrition (TPN), or delivery of all nutrients by intravenous infusion, is usually recommended when oral or tube feedings are expected to be contraindicated or inadequate for more than 5 to 7 days.



Total parenteral nutrition



Parenteral Nutrition

TPN is most commonly used for one of two reasons:

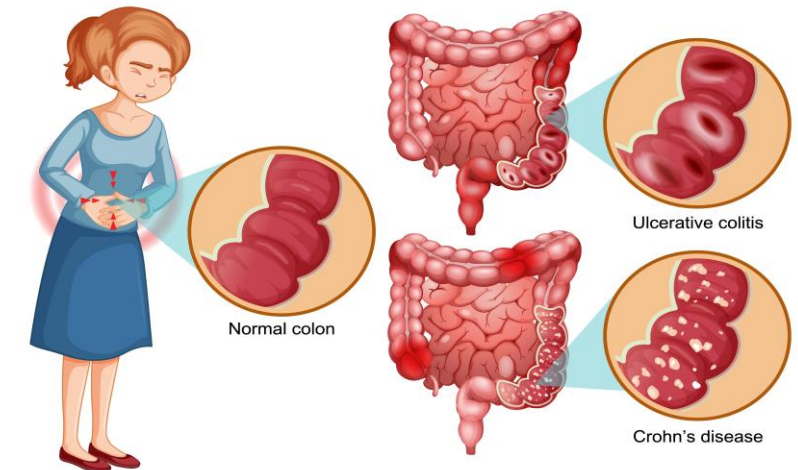
1. **The gastrointestinal (GI) tract is unable to digest or absorb** adequate nutrients. Examples include intractable vomiting, severe diarrhea, prematurely, some cases of abdominal trauma, prolonged ileus, and massive small bowel resection.
2. **There is a need for bowel rest.** Examples include enteral fistulae and acute inflammatory bowel disease that does not respond to other therapies.



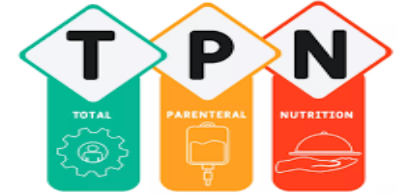
enteral fistulae

Inflammatory Bowel Disease (IBD)

Inflammatory Bowel Disease (IBD) is a group of chronic inflammatory conditions that affect the digestive tract, including Crohn's disease and ulcerative colitis.



TPN is contraindicated



- When the gastrointestinal tract is functional.
- Feeding with TPN alone is associated with atrophy of the intestinal villi and impaired absorption when feedings are resumed
- The intestinal mucosa's ability to act as a barrier to microorganisms is impaired
- Animal studies there is **increased risk of translocating or movement of bacteria through the intestinal wall into the bloodstream and to other organs.** When the patient can tolerate them, small amounts of food help to prevent villous atrophy.

Routes for parenteral Nutrition:

There are several possible routes for TPN delivery:

1) Peripheral Venous Catheters

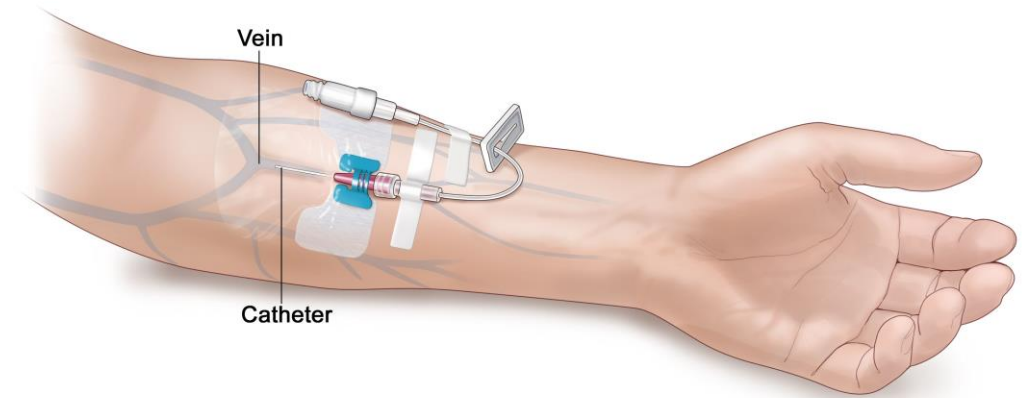
These catheters are usually inserted in the upper extremities.

Advantages: few

Disadvantages

- ☐ Requires good peripheral venous access,
- ☐ Limits its long term use
- ☐ Not usually useful for home TPN.
- ☐ Difficult to deliver adequate kcal to stressed individuals or those with high kcal needs because peripheral veins do not tolerate solution concentration >900 mOsm/kg .
- ☐ Limits dextrose concentrations to about 10%.

Peripheral Venous Catheter



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Routes for parenteral Nutrition:

Central Venous Catheters

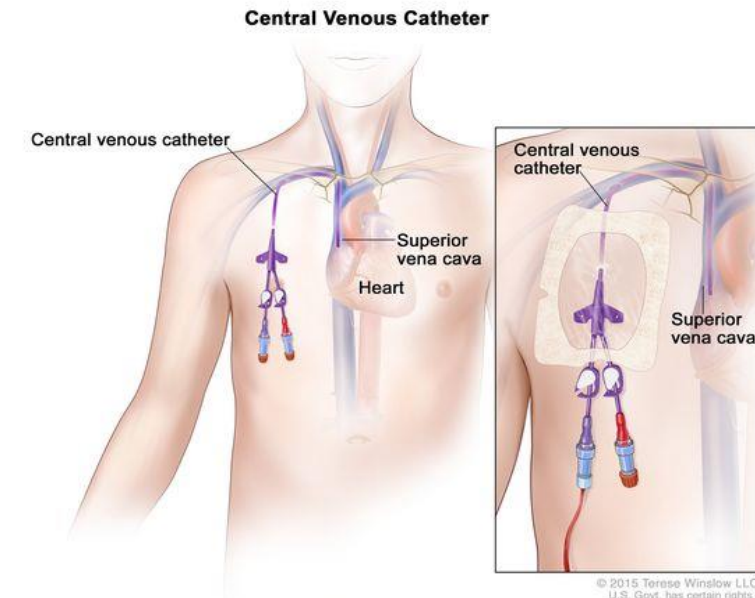
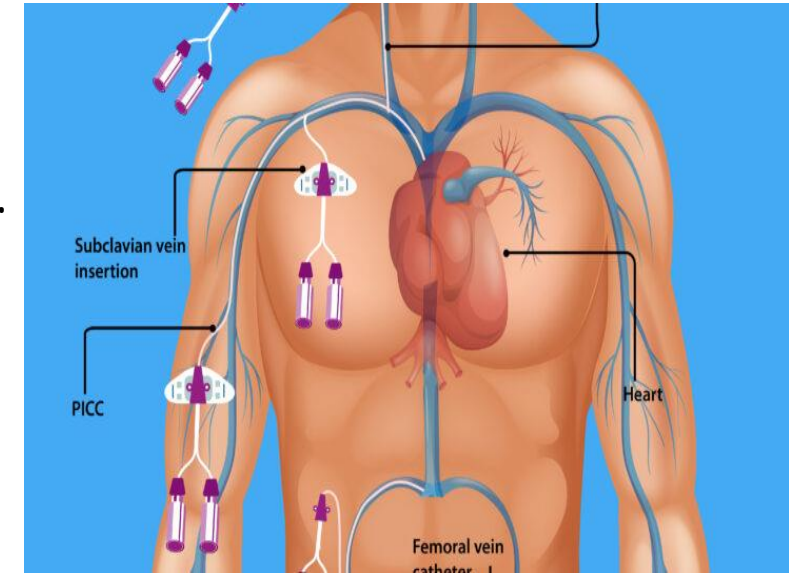
Central venous catheters (CVCs) are usually inserted into the superior or inferior vena cava via the subclavian, internal or external jugular, or femoral veins.

Advantages

- ✓ Allow use of extremely hypertonic solutions (>1800 mOsm/kg)
- ✓ Appropriate for long term and home use.
- ✓ long-term catheters are not easily dislodged because their proximal ends are tunneled under the skin and have cuffs that adhere to the subcutaneous tissues.
- ✓ Multi lumen CVC catheters reduce the risk of drug incompatibilities when more than one medication is given at once and allow for blood sampling via the CVC.

Disadvantages

Complications include pneumothorax, air embolism, central vein thrombosis, catheter-related sepsis, and catheter mobilization



Routes for parenteral Nutrition:

Subcutaneous Infusion Ports

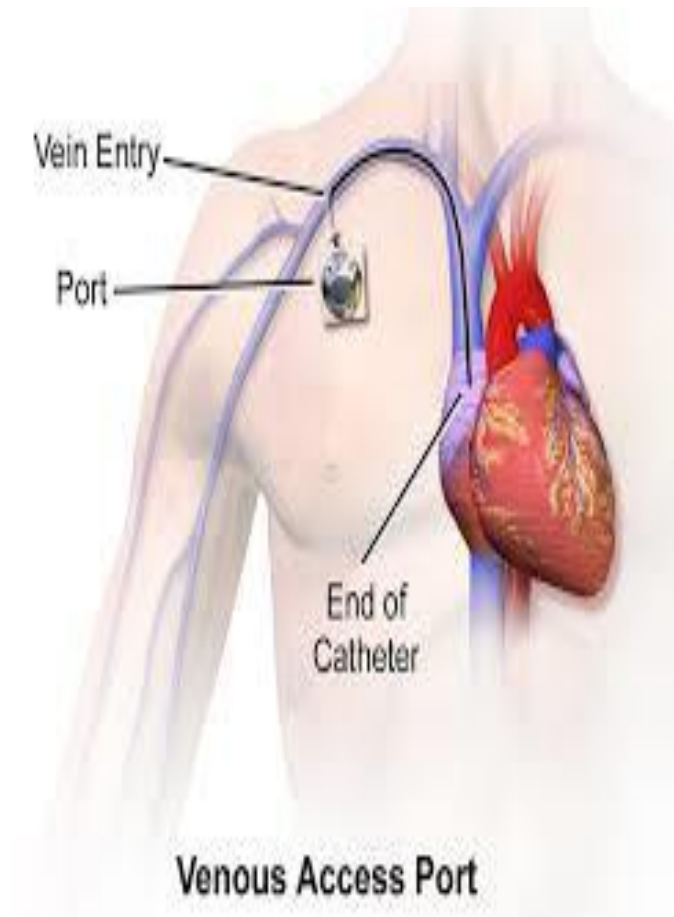
The subcutaneous infusion port is an implanted reservoir with a catheter attached. Before each infusion, a noncoring needle is inserted through the skin and into the septum on the port.

Advantages

Useful for intermittent therapy (e.g., chemotherapy, selected patients needing intermittent TPN); require no flushing or site care when not in use; lower infection rates than standard catheters in several studies.

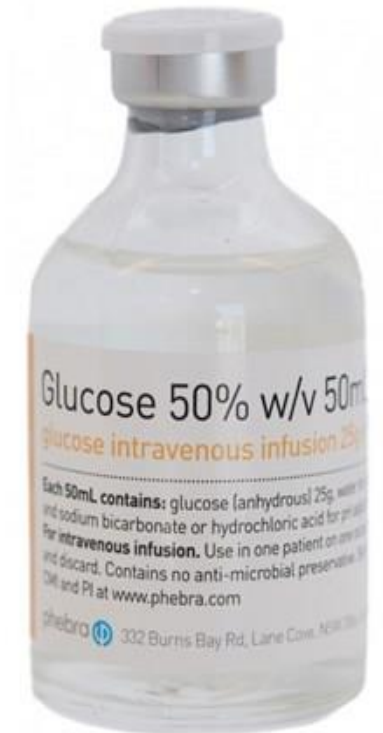
Disadvantages

Requires a needle to access the port; risk of extravasation of infusale if the needle slips out of the port.



Administration regimens and formula:

- Daily requirements are estimated according to **body weight**, and to which are added the **deficit**.
- Daily readjustment depends upon monitoring of **fluid, electrolyte, and nutritional status**.
- **Carbohydrates** are provided mainly by **glucose**, to provide enough energy without overloading the circulation, high concentrations are used, e.g., **20% or 50% glucose**.
- **Proteins** are commonly administered as **L-amino acids solutions** that should contain the essential members. Examples of the commercially available preparations are Vamin, **Freamine**, and Novoplex.
- These solutions also contain **sodium, potassium, phosphate, magnesium, vitamins, trace** elements, and glucose in amounts that vary from one product to the other.



Administration regimens and formula:

- **Fat emulsions** are produced from **Soya oil**, and are available in 10% and 20% concentrations, both are isotonic and can be given via a peripheral vein.
- **Fat emulsions** are good source of energy because liter of the 20% preparation provides 2000Kcal.
- They are usually given **twice weekly** to provide essential fatty acids.
- When mainly using an amino acid/carbohydrate solution, it should be administered in gradually increasing doses over 3-4 days.
- This allows for the development of tolerance.



Preventing and Correcting Complications of Parenteral Nutrition

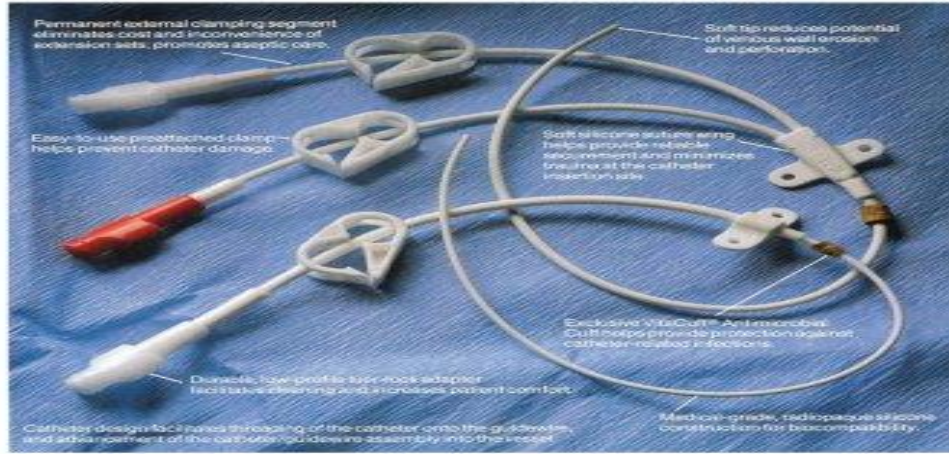
Complication	Sings/symptoms	Prevention/ intervention
Catheter-related sepsis	☐ Fever ☐ Chills ☐ Glucose intolerance ☐ Positive blood culture; bacterial colony counts in blood aspirated from the catheter 5-10 times higher than in blood obtained from a peripheral site	▪ Maintain an intact dressing. ▪ Use aseptic technique whenever handling catheter. ▪ Hang a single bottle of TPN no longer than 24 hr, lipid emulsion no longer than 12 hr; ▪ Salvage of infected catheters has been attempted with an "antibiotic lock" by placing a concentrated antibiotic solution in the catheter for 12 hr.

Preventing and Correcting Complications of Parenteral Nutrition

Complication	Sings/symptoms	Prevention/ intervention
Air embolism	Dyspnea, cyanosis, tachycardia, hypotension, possibly death	○ Use Luer-lok system or secure all connections all. ○ Groshong catheter, which has valve at tip, may reduce risk of air embolism; ○ Use an inline 0.22 µm air-eliminating filter if solutions don't contain lipids or 1.2 µm or larger if solutions contain lipids; ○ Have patient perform Valsalva's maneuver during tubing changes; ○ If air embolism is suggested, place patient in left lateral decubitus position and administer oxygen; immediately notify physician, who will attempt to aspirate air from the heart

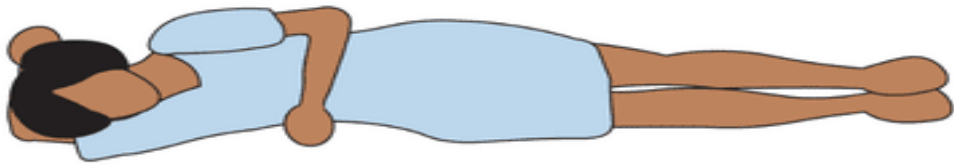
Bard Access Systems Hohn® Central Venous Catheter

The safety of silicone in an intermediate-term percutaneous catheter.

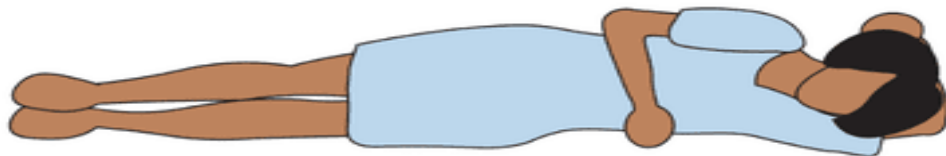


BARD

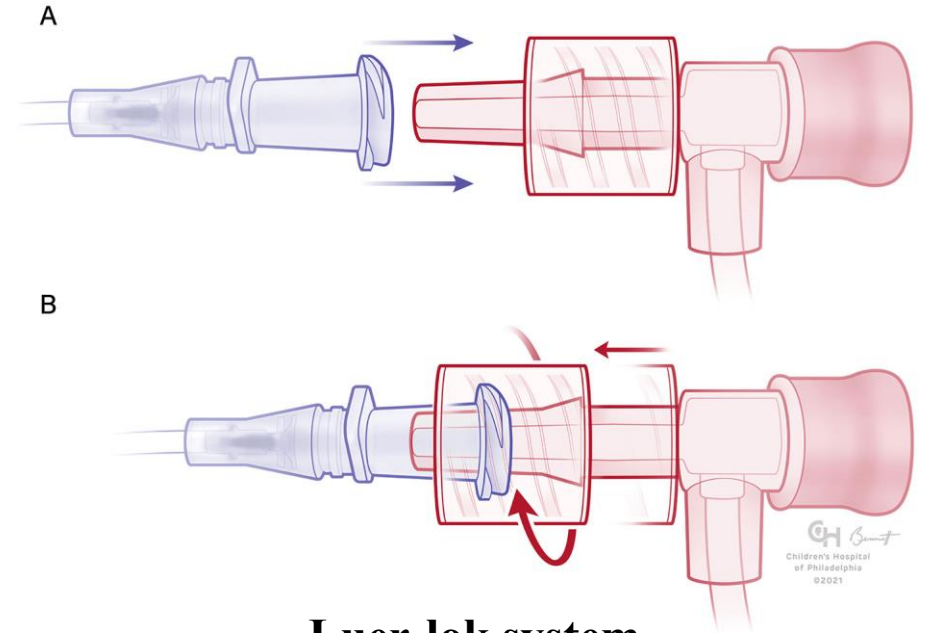
Groshong catheter



Right Lateral Recumbent



Left Lateral Recumbent



Luer-lok system

**Valsalva
Maneuver**



Assessment

Preventing and Correcting Complications of Parenteral Nutrition

Complication	Sings/symptoms	Prevention/ intervention
Central venous thrombosis	Unilateral edema of neck, shoulder, and arm; development of collateral circulation on chest; pain in insertion site	• Follow measures to prevent sepsis; repeated or traumatic catheterizations are most likely to result in thrombosis; • Treatment usually includes anticoagulation; if symptoms are not too severe, the catheter might be left in place; • if symptoms are severe, thrombolytic therapy and/or catheter removal may be necessary.

Preventing and Correcting Complications of Parenteral Nutrition

Complication	Signs/symptoms	Prevention/ intervention
Catheter occlusion or semioclusion	No flow or sluggish flow through the catheter; or infusion through the catheter possible, but blood cannot be aspirated from the catheter	• Flush catheter with heparinized saline if infusion is stopped temporarily; • if catheter appears to be occluded, attempt to aspirate the clot; • Thrombolytic agent may restore patency if clotted or occluded by fibrin sheath; hydrochloric acid, 0.1 N, • Has been used to clear drug precipitates and 70% ethanol to clear lipid precipitates.

Preventing and Correcting Complications of Parenteral Nutrition

Complication	Sings/symptoms	Prevention/ intervention
Hypoglycemia	Diaphoresis, shakiness, confusion, loss of consciousness	• Do not discontinue TPN abruptly, taper rate over several hours; • use pump to regulate infusion so that it remains $\pm 10\%$ of ordered rate; • if hypoglycemia if suggested, then administer oral carbohydrate; • if oral intake is contraindicated or patient is unconscious, a bolus of

Preventing and Correcting Complications of Parenteral Nutrition

Complication	Sings/symptoms	Prevention/ intervention
Hyperglycemia	Thirst, headache, lethargy, increased urination	• Monitor blood glucose frequently until stable; • TPN is usually initiated at a slow rate or With a low dextrose concentration and increased over 2-3 days to avoid hyperglycemia; • the patient may require insulin added to the TPN if the problem is severe

Nursing Role In Nutritional Care of Critically ill Patients

The nurse plays a vital role in initiating, monitoring, and evaluating the response to the nutritional care. Providing nutritional care plan should start with assessment and works through to evaluation applying steps of nursing process.

1-Nutritional Assessment

- **Risk Factors for nutritional deficiencies**
- **Diet History**
- **Physical examination**

2- Dietary Nursing Diagnosis

3- Planning

4- Implementation

5- Evaluation

1-Nutritional Assessment

1) Risk Factors for nutritional deficiencies:

The factors listed below may place a patient at risk for developing, or may denote the presence of nutrient deficiencies:

- Age < 18 years or > 65 years (increased risk age >75 years)
- Recent significant, unintentional weight loss: > 5% in 1 month or >10% in 6 months.
- Excessive alcohol intake, other substance abuse.
- Limited capacity for oral intake (dysphagia, stomatitis, mucositis)
- NPO > 3 days.

1) Risk Factors for nutritional deficiencies:

- Increased metabolic demands: extensive burns, major surgery, trauma, fever, infection, draining, abscesses, wounds, fistulae, pregnancy.
- Protracted nutrient losses: malabsorption syndromes, short gut syndrome, draining abscesses, wounds, fistulae, effusions, renal dialysis.
- Intake of catabolic drugs: corticosteroids, immunosuppressants, antineoplastics.
- Protracted emesis: anorexia nervosa, bulimia, hyperemesis gravidarum, radiation, cancer chemotherapy.
- Chronic disease (especially AIDS, diabetes, cystic fibrosis, stroke, cancer).

Diet History

- Caloric counts are often employed to evaluate and monitor a patient's intake.
- Each day, the nurse accurately records all food and fluids consumed by the patient and specify type and amount.
- The nurse calculates the protein and calories consumed daily and make recommendation for nutritional support.
- It is critical to perform calorie counts during weaning from parenteral nutrition or tube feeding to an oral diet.
- Dietary assessment in ICU also includes assessment of the adequacy of the nutritional support regimen. Data obtained can help to identify actual and potential nutritional problems that classified as primary (caused by poor intake) or secondary (caused by alteration in digestion, absorption, metabolism or excretion of nutrients).

Physical examination:

- Physical examination can provide important indicators of malnourished.
- The most commonly used indices are **height, weight, triceps skin fold thickness, mid arm circumference.**
- In an ideal situation, a measure of body mass index (BMI): weight (kg) divided by height (m) squared is a useful indicator of nutritional state.

$$\text{Body mass index} = \text{Weight (kg)} / \text{Height (m)}^2$$

Physical signs suggestive of nutrient deficiencies

Location	Signs/Symptoms	Deficient Nutrient
hair	lack luster, thin, sparse, easily plucked, dyspigmentation, alternating bands light & dark	energy, protein
face	nasolabial seborrhea, diffuse dyspigmentation paleness	riboflavin iron

Physical signs suggestive of nutrient deficiencies

Location	Signs/Symptoms	Deficient Nutrient
eyes	pale conjunctiva poor dark adaptation, Bitot's spots, corneal or conjunctival xerosis papilledema angular blepharitis intraocular hemorrhage	iron retinol riboflavin, niacin ascorbic acid
lips	bilateral angular stomatitis cheilosis	riboflavin riboflavin, niacin
tongue	scarlet, raw tongue magenta tongue glossitis (beefy red tongue) atrophic filiform papilla (slick tongue)	niacin riboflavin folic acid, cobalamin, iron, niacin, riboflavin folic acid, cobalamin, iron, niacin, riboflavin

Physical signs suggestive of nutrient deficiencies

Location	Signs/Symptoms	Deficient Nutrient
teeth	caries	Fluoride
gums	pale spongy, bleeding, receding, swelling	iron ascorbic acid
endocrine system	thyroid enlargement parotid enlargement	iodine protein
skin	purpura hyperpigmentation desquamatory dermatitis follicular hyperkeratosis, xerosis, mosaic dermatosis petechial hemorrhages thickening/pigmentation at pressure points and sun exposed areas scrotal/vuval dermatosis	vitamin K energy, folic acid, cobalamin, niacin biotin, essential fatty acids, zinc, pyridoxine reinol, essential fatty acids ascorbic acid niacin riboflavin

Physical signs suggestive of nutrient deficiencies

Location	Signs/Symptoms	Deficient Nutrient
nails	koilonychia (spoon nail), paleness	Iron
subcutaneous tissue	fat decreased edema	energy protein, thiamin
musculoskeletal system	wasting craniotables, frontal bossing, knock-kness, beading of ribs, bow legs, musculoskeletal hemorrhage	energy, protein vitamin D ascorbic acid
cardiovascular system	cardiomegaly, tachycardia, CHF, wet beriberi	Thiamin

Physical signs suggestive of nutrient deficiencies

Location	Signs/Symptoms	Deficient Nutrient
nervous system	mental confusion, encephalopathy, calf tenderness, sensory loss, motor weakness, dry beriberi, biolateral absent ankle/knee jerks	thiamin, cobalamin
gastrointestinal system	hepatomegaly	Protein

Laboratory values:

The following tests are:

- ☐ CBC
- ☐ Serum protein measurements: especial Albumin
- ☐ Immunocompetence :The immune system is assessed by determining the TLC and delayed cutaneous hypersensitivity to antigens
- ☐ Urine assays

Dietary Nursing Diagnosis

Possible dietary nursing diagnoses for critically ill patient receiving enteral feeding in relation to nutritional status are:

- High risk for nutritional deficit related to loss of consciousness, protein nutritional deficit related to protein loss.
- Nutritional alteration less than body requirements related to hypermetabolic demands.
- High risk for pulmonary aspiration related to gastric atony
- Decreased gastric motility related to paralytic ileus

Dietary Nursing Diagnosis

As regards to sample nursing diagnoses for critically ill patients requiring nutritional support, they include the following:

- High risk for nutritional deficit related to diarrhea,
- high risk for infection related to catheter sepsis,
- Disturbance in gustatory perception related to lack of oral intake,
- High risk for allergic response related to administration of fat emulsion,
- High risk for trace mineral deficiencies related to dependence on parenteral nutrition.

Based on nursing diagnosis, nutritional plan of care could be conducted, starting with setting priorities and determining patients' goals and then proceed to the step of implementation that

Planning and Implementation

Goals a critically ill patient who is receiving nutritional

- **To ensuring proper delivery of nutrients.**
- **To prevent complications**

When caring for a patient receiving enteral feeding, the nurse must continually assess for the signs and symptoms of complications

Nursing measurements

Respiratory measurement

- Monitoring the degree of gastric retention, by checking for gastric residual every 4 hours or at the beginning of each intermittent feeding, and ensuring that the volume of aspirate does not exceed 50% of the .previous feeding or 50% of the hourly infusion rate for continuous feeding.
- Moreover, aspiration can be prevented by assessment of bowel sounds,
- Keeping the head of the patient's bed elevated 30 to 40 degrees during feeding and half hour after.
- Assessment signs and symptom of respirator aspiration fever, restlessness, tachycardia, tachypnea and a new infiltrate on chest X-ray at which point antibiotic therapy would probably be instituted.
- The nurse must evaluate arterial blood gases (ABGs) daily, as ordered, observing for elevated PaCO₂ level.
- She should also assess the cuff pressures of patients with artificial airways.
- She must ensure that the cuff is inflated to minimum occlusion volume of 20 to 25 cm H₂O.

Cardiovascular, Fluids and Electrolytes measurements

- ❖ The nurse monitors vital signs every 4 hours, an orthostatic drop in the blood pressure of 15 mmHg with increase in pulse of 15 beats indicates a deficit of intravascular water of 15%.
- ❖ The nurse assesses hydration status by measuring of central venous pressure, which provides an accurate picture about fluid volume status of the body.
- ❖ Hydration status can be assessed by other parameters such as daily 4 checking of body weight, where gaining of more than 0.7 kg/day probably indicates fluid overload.
- ❖ Nurses' role is directed toward early detection of water and electrolytes disturbance, through monitoring serum electrolytes, blood glucose, urea nitrogen, creatinine, serum magnesium, phosphorus, calcium, liver function tests and nitrogen balance.
- ❖ Checking urine specific gravity and monitoring blood sugar. Elevation of blood sugar level may increase the risk for osmotic diuresis and fluid volume deficit.

Gastrointestinal measures

- The nurse should assess feeding tube for slippage by recording and Comparing length of tube beyond nares insertion and every 8 hours, marking the feeding tube with permanent ink at its exit site.
- Critical cardiac nurses must be alert to prevent feeding tube obstruction.
- The nurse must assess for delayed gastric emptying by observing for anorexia, nausea and vomiting and by frequent measuring of gastric residuals to determine the patient's ability to tolerate

Gastrointestinal measures

➤ Measure to prevent Diarrhea

- feeding tubes are intermittently
- should be frequently flushed and filled with sterile water to diminish the growth of microorganisms
- The nurse should evaluate patient's tolerance to rate, osmolality and nutrient composition of formula minimize diarrhea
- All personnel handling the feeding should use proper hand washing techniques, Most formulas can be handled at room temperature for 8-12 hours.
- The nurse must record the character, consistency and number of stools per shift.
- The nurse should review the medications that patient is on; evaluating which of them can cause diarrhea or decreased mobility, and ensuring that they are properly diluted.

Metabolic Assessment

The nursing intervention to reduce the incidence of metabolic alteration includes:

- ❑ Repeating nutritional assessment every 7 to 10 days and determining patient's energy expenditure by indirect calorimetry with a metabolic cart or by using an appropriate formula-.
- ❑ The nurse should check calorie and protein count for 5 days, then once weekly, ensuring that patient is receiving electrolytes in the early stage of feeding to prevent rapid, potentially life-threatening cardiac dysrhythmia or respiratory arrest.
- ❑ Moreover, evaluating serum albumin, thyroxine binding, prealbumin, transferrin and retinol binding protein and blood sugar.

Integumentary Assessment

- The nurse should observe patient's skin around tube entry sites for irritation and excoriation.
- Radiological evaluation of the sinuses and otologic examination are indicated.
- The nurse should also assess skin for signs of edema or dehydration.

Psychosocial Assessment

- Ensuring that tube is not uncomfortable or positioned to potentially cause harm.
- Determine if patient has a need to express feelings about changes in body image or inability to eat food orally
- Assess patient's knowledge about the tube feeding and enhancing a patient's tolerance and progress with PN

Evaluation

- The final step in nursing process, is evaluation, through which the nurse determines the extent to which nutritional goal has been achieved.

*Thank
you!*

